Online Retail Sales Analysis

Phase 1: Data Exploration & Quality Assessment

Project Objective:

Perform initial data exploration and data quality assessment on the Online Retail transaction dataset before moving into deeper business analysis.

Dataset: Online Retail Transactions
Total Records: 541,909 transaction rows
Analysis Tool: MySQL
Analyst: Gopi Aneesh Pamulapati

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1. Dataset Structure Overview

Observation:

- The dataset contains **541,909 transaction records**.
- Initial data inspection shows that each row represents an individual product transaction rather than a complete invoice.
- The same InvoiceNo appears multiple times because one invoice can contain multiple products.

Business Meaning:

The dataset follows a transaction-level structure, which allows analysis of customer purchases, product performance, revenue, and order behaviour.

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2. Data Quality Assessment

Missing Value Check

Observation:

No NULL values were found in the dataset columns:

- InvoiceNo
- StockCode
- Description
- Quantity

    - InvoiceDate
    - UnitPrice
    - CustomerID
    - Country

Business Meaning:

The dataset contains complete records and can be used for further analysis without major missing-value handling.

Duplicate Record Check

Observation:

- Duplicate transaction rows were identified where all transaction attributes were identical.
- Repeated InvoiceNo values are expected because invoices contain multiple products.

Business Meaning:

Duplicate checks were performed to distinguish between:

- Valid repeated transactions
- Actual duplicate records

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3. Transaction Validation

Negative Quantity Analysis

Observation:

- Transactions with negative quantities were identified.
- Negative quantities represent returned or cancelled items.

Business Meaning:

Returns are part of normal retail operations and should be analysed separately rather than removed.

Cancelled Invoice Analysis

Observation:

- Cancelled invoices were identified using InvoiceNo values beginning with "C".
- Total cancelled invoices identified: **3,836**
- Total cancelled transaction rows: **9,288**

Business Meaning:

Cancelled orders represent customer returns/refunds and can impact overall revenue calculations.

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4. Date & Time Analysis

Invoice Date Validation

Observation:

- Invoice dates were reviewed to understand transaction coverage.
- The dataset contains transactions from **2001 to 2031 based on stored date values**.
- Date validation was performed to identify possible date inconsistencies.

Business Meaning:

Date fields require careful validation before performing time-based analysis.

Transaction Time Analysis

Observation:

- Invoice timestamps were extracted from InvoiceDate.
- Transaction activity exists across different dates and times.

Business Meaning:

Time-based analysis can be performed later to identify sales trends and customer purchase patterns.

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5. Customer Analysis

Customer Base

Observation:

- Total unique customers identified: **4,373**

Business Meaning:

The dataset contains a large customer base suitable for customer behaviour analysis.

Customer Purchase Behaviour

Observation:

- Customers were analysed based on:
    - Number of transactions
    - Total revenue generated
    - Average spending
    - Purchase period

Key Findings:

- A small group of customers contributes significantly higher revenue compared to others.
- Some customers show high purchasing frequency.

Business Meaning:

Customer segmentation and VIP customer analysis can provide valuable business insights.

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6. Product Analysis

Product Availability

Observation:

- Total unique products identified: **3,958 StockCodes**

Business Meaning: The company has a wide product catalogue suitable for product performance analysis.

Product Sales Performance

Observation:

Products were analysed based on:

- Quantity sold
- Revenue generated
- Customer reach
- Average selling price

Business Meaning:

Product-level analysis helps identify:

- Best-selling products
- High revenue products
- Customer favourite products

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7. Country Analysis

Market Distribution

Observation:

- The dataset contains transactions from **38 countries**.

Key Findings:

- United Kingdom has the highest:
    - Transaction volume
    - Revenue contribution
    - Customer base

Business Meaning:

The company has strong dependence on the UK market, while other countries represent expansion opportunities.

8. Country Revenue Analysis

Observation:

Countries were compared based on:

- Total revenue
- Number of invoices
- Number of customers
- Average transaction value

Business Meaning:

Different markets show different purchasing behaviours, which can support international growth strategies.

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9. Product Return Analysis

Returned Products

Observation:

- Returned quantities were analysed using cancelled invoices.
- Products with higher return volumes were identified.

Business Meaning:

Return behaviour can help identify:

- Product quality issues
- Customer dissatisfaction
- Inventory improvement opportunities

Phase 1 Summary

During Phase 1, the dataset was successfully explored and validated.

Completed Analysis:

- Dataset structure review

- Missing value assessment
- Duplicate transaction checks
- Negative quantity analysis
- Cancelled order identification
- Date validation
- Customer analysis
- Product analysis
- Country analysis
- Return analysis

Conclusion:

The next phase will focus on Business Analysis, where customer behaviour, product performance, country performance, and sales trends will be analysed to identify meaningful business insights.

Phase 2: Business Analysis & Sales Performance Insights

Project Objective:

Perform business analysis on the Online Retail transaction dataset to identify customer purchasing behaviour, product performance, country performance, and sales trends using SQL.

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1. Customer Behaviour Analysis

Customer Revenue Analysis

Observation:

• Customers were analysed based on total revenue generated using transaction value (Quantity × UnitPrice).
• High-value customers were identified based on their total spending.

Key Findings:

• A small group of customers contributes significantly higher revenue compared to other customers.
• CustomerID 14646 generated the highest revenue among customers analysed.

Business Meaning:

High-value customers play an important role in revenue generation. Identifying these customers helps businesses focus on customer retention and loyalty strategies.

Customer Purchase Frequency Analysis

Observation:

• Customers were analysed based on the number of unique invoices generated.
• Purchase frequency was calculated using distinct InvoiceNo values.

Key Findings:

• Some customers placed significantly more orders compared to others.
• A few customers showed repeated purchasing behaviour.

Business Meaning:

Frequent customers represent valuable repeat buyers and can be targeted through retention programs and personalised offers.

Repeat Customer Analysis

Observation:

• Customers were analysed based on multiple purchases using their invoice history.
• Customers with more than one invoice were identified as repeat customers.

Key Findings:

• A large number of customers made repeat purchases, especially in the United Kingdom market.
• Repeat customers contribute significantly to business activity.

Business Meaning:

Customer retention is important because repeat customers provide continuous revenue and long-term business value.

Average Order Value Analysis

Observation:

• Average order value was calculated by analysing revenue generated per customer order.

Key Findings:

• Some customers have higher average spending per order compared to others.
• High average order value customers were identified.

Business Meaning:

Customers with higher spending patterns can be targeted for premium products and personalised marketing strategies.

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2. Product Performance Analysis

Product Revenue Analysis

Observation:

• Products were analysed based on total revenue generated using Quantity × UnitPrice.

Key Findings:

• Certain products contribute significantly higher revenue compared to other products.
• Revenue contribution is concentrated among a smaller group of products.

Business Meaning:

Understanding top revenue-generating products helps businesses focus on inventory planning and product promotion.

Product Quantity Analysis

Observation:

• Products were analysed based on total quantity sold.

Key Findings:

• Some products have high customer demand based on sales volume.
• High-volume products may not always be the highest revenue products.

Business Meaning:

Both sales volume and revenue should be considered when making inventory and product decisions.

Product Customer Reach Analysis

Observation:

• Products were analysed based on the number of unique customers purchasing them.

Key Findings:

• Some products are purchased by a wider customer base compared to others.

Business Meaning:

Products with high customer reach indicate strong customer acceptance and market demand.

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3. Country Performance Analysis

Country Revenue Analysis

Observation:

• Countries were compared based on total revenue generated.

Key Findings:

• The United Kingdom generated the highest revenue contribution.
• Other European countries contributed additional revenue streams.

Business Meaning:

The business is highly dependent on the UK market, while international markets provide opportunities for further growth.

Country Order Analysis

Observation:

• Countries were analysed based on the number of unique invoices.

Key Findings:

• The United Kingdom recorded the highest order volume.
• Germany and France were among the other active markets.

Business Meaning:

Order volume helps identify the strongest customer markets and supports regional sales planning.

Country Customer Analysis

Observation:

• Countries were compared based on the number of unique customers.

Key Findings:

• The United Kingdom has the largest customer base.
• Other countries have smaller but valuable customer groups.

Business Meaning:

Customer distribution helps identify important markets and potential expansion opportunities.

Average Order Value by Country

Observation:

• Countries were analysed based on average transaction value.

Key Findings:

• Some countries have fewer orders but higher average spending per order.

Business Meaning:

Different markets show different purchasing behaviours, which can support targeted international strategies.

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4. Time-Based Sales Analysis

Yearly Revenue Trend Analysis

Observation:

• Revenue trends were analysed across different years using InvoiceDate.

Key Findings:

• Sales performance varies across different years.
• Certain years generated higher revenue compared to others.

Business Meaning:

Understanding yearly trends helps identify business growth patterns and changes in sales performance.

Monthly Sales Trend Analysis

Observation:

• Monthly revenue and order trends were analysed using InvoiceDate.

Key Findings:

• Sales activity changes across different months.
• Certain months show higher transaction activity.

Business Meaning:

Seasonal sales patterns can help businesses improve inventory planning and marketing campaigns.

Phase 2 Summary

During Phase 2, business analysis was performed using SQL to understand customer behaviour, product performance, market performance, and sales trends.

Completed Analysis:

• Customer revenue analysis
• Customer purchase frequency analysis
• Repeat customer analysis
• Average order value analysis
• Product revenue analysis
• Product demand analysis
• Country revenue and order analysis
• Customer distribution by country
• Monthly and yearly sales trend analysis

Conclusion:

The analysis identified key customers, high-performing products, important markets, and sales patterns within the Online Retail dataset. These insights provide a foundation for further data cleaning and final business analysis.

Phase 3: Data Cleaning & Final Dataset Preparation

Project Objective:

Perform data cleaning and final dataset preparation on the Online Retail transaction dataset by removing duplicate records, validating cleaned data quality, and creating a final analysis-ready table for exploratory data analysis.

Initial Cleaned Records: 396,485 transaction rows
Final Analysis Records: 391,298 transaction rows

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1. Cleaned Dataset Preparation

Retail_Cleaned Table Creation

Observation:

• A cleaned dataset table (Retail_Cleaned) was created after performing initial data quality checks.

• The purpose of the cleaned table was to:

- Preserve the original raw dataset.
- Store processed transaction records.
- Prepare the dataset for further validation and analysis.

Business Meaning:

Creating a separate cleaned table ensures that the original data remains unchanged while providing a controlled environment for analysis.

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2. Duplicate Transaction Analysis

Exact Duplicate Record Identification

Observation:

• Duplicate transaction records were identified by comparing all transaction attributes:

- InvoiceNo

- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

• Duplicate analysis was performed to differentiate between:

- Valid repeated invoice transactions.
- Actual identical duplicate records.

Key Findings:

• Exact duplicate transaction rows were identified.

• Total duplicate rows that needed removal:

5,187 rows

Business Meaning:

Duplicate records can create inaccurate business calculations, especially when analysing:

- Total revenue
- Product sales
- Invoice values
- Customer purchase behaviour

Removing duplicate transactions improves the reliability of analysis results.

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3. Final Analysis Table Creation

Retail_Final Table Creation

Observation:

• A final analysis table (Retail_Final) was created using distinct transaction records from the cleaned dataset.

• Duplicate rows were removed using the DISTINCT function.

Process:

Retail_Cleaned

↓

SELECT DISTINCT

↓

Retail_Final

Key Findings:

Before duplicate removal:

• Retail_Cleaned records:

396,485 rows

After duplicate removal:

• Retail_Final records:

391,298 rows

Total records removed:

5,187 duplicate rows

Business Meaning:

The final dataset contains unique transaction records and can be used confidently for exploratory data analysis and business reporting.

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4. Duplicate Removal Validation

Final Duplicate Check

Observation:

• Duplicate validation was performed on Retail_Final.

• The validation included checking identical combinations of:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

Key Findings:

• No duplicate transaction records were found.

Result:

0 duplicate rows returned

Business Meaning:

The final dataset successfully passed duplicate validation and is ready for analysis.

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5. Cancelled Invoice Validation

Cancelled Transaction Check

Observation:

• Cancelled invoices were checked in the final dataset using InvoiceNo values beginning with "C".

Validation Method:

LEFT(InvoiceNo,1) = 'C'

Key Findings:

Raw dataset:

• Cancelled transaction rows identified:

9,288 rows

Final dataset:

• Cancelled invoices remaining:

0 rows

Business Meaning:

Cancelled transactions were successfully removed from the final analysis dataset, preventing inaccurate revenue and sales performance calculations.

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6. Final Dataset Quality Assessment

Retail_Final Data Quality Status

Observation:

The final analysis table was validated for:

• Duplicate records
• Cancelled invoices
• Data consistency

Final Dataset Status:

Quality Check	Result
Duplicate records	Removed
Exact duplicate rows remaining	0
Cancelled invoices remaining	0
Final records	391,298

Business Meaning:

The final dataset is clean, consistent, and suitable for performing exploratory data analysis and generating business insights.

Phase 3 Summary

During Phase 3, the Online Retail dataset was transformed into a final analysis-ready dataset.

Completed Activities:

- Created cleaned transaction table
- Identified exact duplicate records
- Removed duplicate transaction rows
- Created final analysis table (Retail_Final)
- Validated duplicate removal
- Validated cancelled invoice removal
- Prepared dataset for EDA

Conclusion:

The data cleaning process successfully improved dataset quality by removing duplicate and cancelled transactions. The final dataset contains **391,298 unique transaction records** and is ready for the next phase: **Exploratory Data Analysis (EDA)**.

Phase 4: Exploratory Data Analysis (EDA)

Project Objective

Perform exploratory data analysis on the cleaned **Retail_Final** dataset to identify sales patterns, customer purchasing behaviour, product performance, revenue distribution, and market performance through SQL-based business analysis.

Analysis Dataset: Retail_Final

Final Records Analysed: 391,298 Transaction Records

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1. Sales Performance Analysis

Total Revenue Analysis

Observation

The total sales revenue was calculated using:

Revenue = Quantity × UnitPrice

Cancelled orders and duplicate records were excluded from the final dataset before analysis.

Business Meaning

Revenue analysis provides an overview of overall business performance and forms the foundation for evaluating customers, products, countries, and sales trends.

Highest Revenue Orders

Observation

Individual invoices were analysed to identify the highest-value customer orders.

Key Findings

- Several invoices generated exceptionally high revenue compared with the average order value.
- High-value invoices contributed significantly to total business revenue.

Business Meaning

Large-value invoices indicate bulk purchases and valuable business customers.

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2. Customer Analysis

Highest Lifetime Revenue Customers

Observation

Customers were analysed based on their total lifetime spending across all completed transactions.

Key Findings

Top customers generated substantially higher revenue than the remaining customer base.

Example:

Customer ID	Lifetime Revenue
14646	279,138.02
18102	259,657.30
17450	194,390.79

Business Meaning

A relatively small number of customers contribute a significant proportion of overall revenue.

Repeat Customer Analysis

Observation

Customer purchase frequency was analysed using distinct Invoice Numbers.

Key Findings

- Repeat customers generated significantly more revenue than one-time customers.
- Customer loyalty plays an important role in business performance.

Business Meaning

Customer retention strategies should remain a major business priority.

Revenue Contribution by Top Customers

Observation

The contribution of the top 10 highest-spending customers was analysed.

Key Findings

Top 10 customers generated:

17.44% of total company revenue

Business Meaning

A small customer segment contributes a disproportionately large share of business revenue.

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3. Product Performance Analysis

Best-Selling Products

Observation

Products were ranked according to total quantity sold.

Key Findings

Several products consistently recorded very high sales volume.

Business Meaning

These products represent strong customer demand and require efficient inventory management.

Highest Revenue Products

Observation

Products were ranked according to total revenue generated.

Key Findings

A limited number of products generated the majority of product revenue.

Business Meaning

High-revenue products should receive greater promotional focus and inventory availability.

Revenue Per Customer

Observation

Revenue generated per unique customer for each product was calculated.

Key Findings

Some products generated very high revenue despite being purchased by relatively few customers.

Business Meaning

These products represent premium-value items with higher customer spending.

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4. Country Analysis

Revenue by Country

Observation

Revenue generated by each country was analysed.

Key Findings

The United Kingdom generated the highest total revenue.

Other major contributing countries included:

- Germany
- France
- Netherlands
- Australia
- EIRE

Business Meaning

The business remains highly dependent on the UK market while international markets present growth opportunities.

Customer Distribution

Observation

Customer distribution was analysed across countries.

Key Findings

The United Kingdom had the largest customer base.

Business Meaning

Future expansion strategies may focus on increasing customer acquisition outside the UK.

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5. Sales Trend Analysis

Monthly Revenue Trend (2011)

Observation

Monthly revenue was analysed using InvoiceDate.

Key Findings

Revenue increased steadily throughout 2011 before declining during December.

Highest Revenue Month:

November 2011

Lowest Revenue Month:

February 2011

Business Meaning

Seasonal purchasing behaviour influences overall company performance.

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6. Revenue Distribution Analysis

Revenue by Customer Segment

Observation

Customers were classified into repeat and one-time customers.

Key Findings

Repeat customers contributed substantially more revenue than one-time customers.

Business Meaning

Customer loyalty programs can significantly improve long-term profitability.

Revenue Concentration

Observation

Revenue distribution was analysed across products and customers.

Key Findings

Revenue is concentrated among:

- A small group of customers
- A limited number of products

Business Meaning

The business follows a Pareto-like revenue distribution where a minority of entities generate the majority of revenue.

Phase 4 Summary

During Phase 4, exploratory data analysis was successfully performed on the cleaned Retail_Final dataset.

Completed Analysis

- Total Revenue Analysis
- Highest Revenue Customers
- Customer Lifetime Revenue
- Repeat Customer Analysis
- Revenue Contribution by Top Customers
- Best-Selling Products
- Highest Revenue Products
- Revenue Per Customer
- Country Revenue Analysis
- Customer Distribution by Country
- Monthly Revenue Trend Analysis
- Revenue Distribution Analysis

Conclusion

The exploratory data analysis identified important business patterns regarding customer behaviour, product performance, market contribution, and sales trends. These findings provide valuable insights that support inventory planning, customer retention strategies, product optimisation, and international business growth.

Phase 5: Business Insights & Recommendations

Project Objective

Summarise the key findings from the exploratory data analysis and provide business recommendations that can support strategic decision-making, improve customer retention, optimise inventory management, and increase overall business profitability.

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1. Key Business Insights

Insight 1: Revenue is highly concentrated among a small group of customers

Observation

The analysis showed that the **top 10 customers generated 17.44% of the total company revenue**, indicating that a relatively small customer segment contributes significantly to overall sales.

Business Impact

The business depends heavily on a limited number of high-value customers.

Recommendation

Develop customer loyalty programs, personalised promotions, and premium services to retain these valuable customers.

Insight 2: Repeat customers generate substantially higher revenue

Observation

Customers with multiple purchases contributed significantly more revenue than one-time customers.

Business Impact

Customer retention has a greater impact on profitability than acquiring new customers alone.

Recommendation

Introduce loyalty rewards, targeted email campaigns, and personalised product recommendations to encourage repeat purchases.

Insight 3: Revenue is concentrated among a limited number of products

Observation

Only a small number of products generated the majority of overall revenue.

Business Impact

Certain products are key revenue drivers and should receive greater business attention.

Recommendation

Maintain sufficient inventory levels, promote high-performing products, and reduce stock shortages.

Insight 4: High sales volume does not always mean high revenue

Observation

Several products recorded high sales quantities but generated comparatively lower revenue.

Business Impact

Product profitability should not be measured solely by sales volume.

Recommendation

Evaluate product pricing and profit margins before making inventory decisions.

Insight 5: Premium products generate high revenue despite lower customer reach

Observation

Some products generated high revenue while being purchased by relatively few customers.

Business Impact

Premium products represent valuable opportunities for increasing profit margins.

Recommendation

Target premium customers through personalised marketing campaigns and exclusive offers.

Insight 6: The United Kingdom dominates business performance

Observation

The United Kingdom generated the highest revenue, customer count, and transaction volume among all countries analysed.

Business Impact

The company relies heavily on a single market.

Recommendation

Diversify revenue sources by expanding marketing efforts in high-potential international markets such as Germany, France, Australia, and the Netherlands.

Insight 7: Sales demonstrate clear seasonal trends

Observation

Monthly revenue analysis showed that sales increased steadily throughout 2011 before reaching the highest revenue during **November**.

Business Impact

Seasonal demand significantly influences business performance.

Recommendation

Increase inventory and promotional campaigns before peak sales periods while planning inventory carefully during slower months.

Insight 8: Data quality significantly improved after cleaning

Observation

The data cleaning process removed **5,187 duplicate transaction records** and excluded cancelled transactions, producing a final analysis dataset containing **391,298 unique records**.

Business Impact

Accurate reporting depends on high-quality data.

Recommendation

Implement automated data validation procedures to minimise duplicate records and improve future reporting accuracy.

Insight 9: Customer spending patterns vary considerably

Observation

Average order value differs significantly across customers and countries.

Business Impact

Different customer groups require different pricing and marketing strategies.

Recommendation

Segment customers according to purchasing behaviour and develop targeted promotional strategies.

Insight 10: SQL-driven analytics supports business decision-making

Observation

The project successfully transformed raw transaction data into meaningful business insights through SQL-based analysis.

Business Impact

Data-driven decision-making improves customer retention, product optimisation, inventory planning, and market expansion.

Recommendation

Develop interactive dashboards using **Power BI** or **Tableau** to allow management to monitor these KPIs continuously.

During Phase 5, the analytical findings were translated into actionable business insights and recommendations. The project demonstrated how SQL can be used not only for querying data but also for solving real business problems by identifying revenue drivers, customer behaviour, product performance, market opportunities, and seasonal sales trends.

Final Project Conclusion

This project successfully analysed the Online Retail transaction dataset using SQL through five structured phases: **Data Exploration, Business Analysis, Data Cleaning, Exploratory Data Analysis, and Business Insights & Recommendations**. The analysis identified high-value customers, top-performing products, country-level sales performance, customer purchasing behaviour, and seasonal revenue trends. By converting raw transactional data into meaningful business insights, the project demonstrates practical SQL skills and a business-oriented approach to data analysis, making it suitable for a professional **Data Analyst portfolio**.